

NPRG Formalism for Navier–Stokes Turbulence: Symmetries, Ansatz, and Flow Equations

Note on notation. Greek indices $\alpha, \beta, \gamma, \dots$ and Latin indices $i, j, d, b, c, e, f, \dots$ label Cartesian components. $u_\alpha, \bar{u}_\alpha, p, \bar{p}$ are the velocity, response-velocity, pressure, and response-pressure fields; Γ_κ is the scale-dependent effective average action; R_κ and N_κ are the regulator and forcing kernels; $P_{\alpha\beta}^\perp(p) = \delta_{\alpha\beta} - p_\alpha p_\beta / p^2$ is the transverse projector. All equations below are numbered sequentially in the order they are introduced.

1 Symmetries of the Navier–Stokes Effective Action

The Navier–Stokes action $S \equiv S_0 + S_\kappa$ admits three gauge symmetries: (i) invariance under gauged shifts of the pressure fields, (ii) time-gauged Galilean symmetry, (iii) invariance under a time-gauged shift of the response fields.

1.1 Time-gauged Galilean transformation

$$\begin{aligned}\delta u_\alpha(x) &= -\dot{\epsilon}_\alpha(t) + \epsilon_\beta(t) \partial_\beta u_\alpha(x), \\ \delta \bar{u}_\alpha(x) &= \epsilon_\beta(t) \partial_\beta \bar{u}_\alpha(x), \\ \delta p(x) &= \epsilon_\beta(t) \partial_\beta p(x), \\ \delta \bar{p}(x) &= \epsilon_\beta(t) \partial_\beta \bar{p}(x),\end{aligned}\tag{1}$$

with $\dot{\epsilon}_\alpha \equiv \partial_t \epsilon_\alpha$. A constant $\epsilon(t) \equiv \epsilon$ generates an ordinary space translation, while $\epsilon(t) \equiv \epsilon t$ generates the usual (nongauged) Galilean boost.

1.2 Time-gauged shift transformation

$$\begin{aligned}\delta \bar{u}_\alpha(x) &= \bar{\epsilon}_\alpha(t), \\ \delta \bar{p}(x) &= u_\beta(x) \bar{\epsilon}_\beta(t).\end{aligned}\tag{2}$$

1.3 Ward identity from the pressure sector

$$\frac{\delta \Gamma_\kappa}{\delta p(x)} = \frac{\delta S_0}{\delta p(x)}, \quad \frac{\delta \Gamma_\kappa}{\delta \bar{p}(x)} = \frac{\delta S_0}{\delta \bar{p}(x)}.\tag{3}$$

1.4 Variation under the Galilean transformation

The Navier–Stokes action is invariant under the transformation above except for the term containing the material (Lagrangian) time derivative $D_t u_\alpha(x) \equiv \partial_t u_\alpha(x) + u_\beta(x) \partial_\beta u_\alpha(x)$, whose variation is

$$\delta \int_x \bar{u}_\alpha(x) D_t u_\alpha(x) \equiv \delta S = - \int_x \ddot{\epsilon}_\alpha(t) \bar{u}_\alpha(x). \quad (4)$$

1.5 Ward identity for the Galilean symmetry

$$\int_x \left\{ [\delta_{\alpha\beta} \partial_t + \partial_\beta u_\alpha(x)] \frac{\delta \Gamma_\kappa}{\delta u_\alpha(x)} + \partial_\beta \bar{u}_\alpha(x) \frac{\delta \Gamma_\kappa}{\delta \bar{u}_\alpha(x)} + \partial_\beta p(x) \frac{\delta \Gamma_\kappa}{\delta p(x)} + \partial_\beta \bar{p}(x) \frac{\delta \Gamma_\kappa}{\delta \bar{p}(x)} \right\} = - \int_x \partial_t^2 \bar{u}_\beta(x). \quad (5)$$

1.6 Variation under the shift transformation

$$\delta S = - \int_x \dot{\epsilon}_\beta(t) u_\beta(x). \quad (6)$$

1.7 Ward identity for the shift symmetry

$$\int_x \left\{ \frac{\delta \Gamma_\kappa}{\delta \bar{u}_\alpha(x)} + u_\alpha(x) \frac{\delta \Gamma_\kappa}{\delta \bar{p}(x)} \right\} = \int_x \partial_t u_\alpha(x). \quad (7)$$

1.8 General structure of the effective action

$$\Gamma_\kappa[u, \bar{u}, p, \bar{p}] = \int_x \left\{ \bar{u}_\alpha \left(\partial_t u_\alpha + \lambda u_\beta \partial_\beta u_\alpha + \frac{\partial_\alpha p}{\rho} \right) + \bar{p} \partial_\alpha u_\alpha \right\} + \tilde{\Gamma}_\kappa[u, \bar{u}], \quad (8)$$

where the explicit terms keep their bare form (not renormalized), the functional $\tilde{\Gamma}_\kappa$ is invariant under both the time-gauged Galilean and shift transformations, and the coupling λ (multiplying the advective nonlinearity) is likewise not renormalized.

2 Leading-Order (LO) Ansatz

At leading order, $\tilde{\Gamma}_\kappa$ is truncated at quadratic order in the response field $\bar{u}$, and the (nonbare) frequency dependence is neglected:

$$\tilde{\Gamma}_\kappa[u, \bar{u}] = \int_{t,x,x'} \left\{ \bar{u}_\alpha(t, x) f_{\kappa, \alpha\beta}^\nu(x - x') u_\beta(t, x') - \bar{u}_\alpha(t, x) f_{\kappa, \alpha\beta}^D(x - x') \bar{u}_\beta(t, x') \right\}. \quad (9)$$

Here f_κ^ν and f_κ^D are the running (viscosity- and noise-type) functions of the ansatz.

3 General Form of the Regulator/Running Functions f_κ^X

Allowing the running functions f_κ^X ($X = \nu, D$) to retain an arbitrary dependence on the (covariant) material time derivative $\tilde{D}_t$, in addition to their momentum dependence through $-\nabla^2$, one writes the general operator expansion

$$f_\kappa^X(\tilde{D}_t, -\nabla^2) = \sum_{m,n=0}^{\infty} a_{\kappa,mn}^X (\tilde{D}_t)^m (-\nabla^2)^n, \quad (10)$$

where the covariant material derivative is

$$\tilde{D}_t = \partial_t + u^i \nabla_i, \quad (11)$$

with ∇_i the (spatial) covariant derivative built from the flow itself, reducing to $\tilde{D}_t = \partial_t + u^i \partial_i$ in flat space. This generalizes the bare Navier–Stokes material derivative D_t to an operator that can appear to arbitrary order m, n inside the running functions f_κ^X of the ansatz above.

4 Wetterich Equation

The exact NPRG flow of the effective average action Γ_κ is given by the Wetterich equation, which in Fourier space reads

$$\partial_\kappa \Gamma_\kappa = \frac{1}{2} \text{Tr} \int_q \partial_\kappa R_\kappa(q) \cdot G_\kappa(q), \quad (12)$$

where $R_\kappa(q)$ is the Fourier transform of the regulator matrix, and

$$G_\kappa \equiv [\Gamma_\kappa^{(2)} + R_\kappa]^{-1} \quad (13)$$

is the full (field-dependent), scale- κ renormalized propagator matrix of the theory.

5 Trace Coefficients for the (1,1) and (2,0) Flow Equations

5.1 Preliminaries: propagators and threshold functions

5.1.1 Regularized propagators

$$\begin{aligned} G_{\alpha\beta}^{\mu\bar{\mu}}(\omega, q) &= P_{\alpha\beta}^\perp(q) \frac{1}{\Gamma_\perp^{(1,1)}(-\omega, q) + R_\kappa(q)}, \\ G_{\alpha\beta}^{\mu\mu}(\omega, q) &= -P_{\alpha\beta}^\perp(q) \frac{\Gamma_\perp^{(0,2)}(\omega, q) - 2N_\kappa(q)}{[\Gamma_\perp^{(1,1)}(\omega, q) + R_\kappa(q)]^2}. \end{aligned} \quad (14)$$

5.1.2 Shorthand notation

For the $\mu\bar{\mu}$ sector:

$$\bar{G} \equiv G^{\mu\bar{\mu}}(-\omega, q), \quad \bar{G}' \equiv G^{\mu\bar{\mu}}(-\omega_1 - \omega_2, p + q), \quad G' \equiv G^{\mu\bar{\mu}}(\omega_1 + \omega_2, p + q).$$

For the $\mu\mu$ sector:

$$\bar{G} \equiv G^{\mu\mu}(-\omega, q), \quad \bar{G}' \equiv G^{\mu\mu}(-\omega_1 - \omega_2, p + q), \quad G' \equiv G^{\mu\mu}(\omega_1 + \omega_2, p + q).$$

5.1.3 Threshold functions

$$A_v(p, \omega_1, \omega_2) = \frac{f_v(p, \omega_1 + \omega_2)}{\omega_1 + \omega_2} + \frac{f_v(p, \omega_1)}{\omega_1}, \quad (15)$$

$$A_d(p, \omega_1, \omega_2) = f_d(p, \omega_1) + f_d(p, \omega_2), \quad (16)$$

$$D_v(p, \omega_1, \omega_2, \omega_3) = \frac{f_v(p, \omega_1 + \omega_2 + \omega_3)}{(\omega_1 + \omega_2 + \omega_3)^2} + \frac{f_v(p, \omega_1 + \omega_2)}{(\omega_1 + \omega_2)^2} + \frac{f_v(p, \omega_1)}{\omega_1^2}, \quad (17)$$

$$D_d(p, \omega_1, \omega_2, \omega_3) = \frac{f_d(p, \omega_1 + \omega_2 + \omega_3)}{(\omega_1 + \omega_2 + \omega_3)^2} + \frac{f_d(p, \omega_1 + \omega_2)}{(\omega_1 + \omega_2)^2} + \frac{f_d(p, \omega_1)}{\omega_1^2}. \quad (18)$$

5.2 Equations for (1,1)

5.2.1 Structure of the traces

$$\begin{aligned} \text{Tr}[\Gamma^{(4)}] &= c_1 \cdot \left\{ \partial_\kappa R_\kappa(q) [G^{\mu\bar{\mu}}(-w, q) G^{\mu\mu}(w, q) + G^{\mu\mu}(w, q) G^{\mu\bar{\mu}}(w, q)] \right\} \\ &+ c_2 \cdot \left\{ \partial_\kappa R_\kappa(q) [G^{\mu\bar{\mu}}(-w, q)^2 + G^{\mu\bar{\mu}}(w, q)^2] \right\}, \end{aligned} \quad (19)$$

$$\begin{aligned} \text{Tr}[\Gamma^{(3)}] &= \partial_\kappa R_\kappa \left\{ c_1 [\bar{G}'_{\mu\bar{\mu}} (G^{\mu\mu} \bar{G}^{\mu\bar{\mu}} + G^{\mu\bar{\mu}} G^{\mu\mu}) + G'_{\mu\mu} (\bar{G}^{\mu\bar{\mu}})^2] \right. \\ &\quad \left. + c_2 [G'_{\mu\bar{\mu}} (\bar{G}^{\mu\bar{\mu}})^2 + (G^{\mu\bar{\mu}})^2] + c_3 [G'_{\mu\bar{\mu}} (G^{\mu\bar{\mu}})^2] \right\}. \end{aligned} \quad (20)$$

Note: the coefficients c_1, c_2, c_3 appearing in the three-point trace above are built from the two building blocks c' and c'' defined below; they are distinct from the c_1, c_2 of the four-point trace, given immediately above it.

5.2.2 Coefficients c_1, c_2 in $\text{Tr}[\Gamma^{(4)}]$

$$\begin{aligned} c_1 \Rightarrow & p^2 P_{ij}^\perp(p) P_{bc}^\perp(q) \left\{ (p)_c (p+q)_d P_{ij}^\perp(q) D_v(-q, w_1, w_2, -w_2) \right. \\ & + (p)_d (p-q)_c P_{ij}(q) D_v(q, w_1, -w_2, w_2) \\ & + (-q)_i (p-q)_c P_{jd}(q) D_v(q, -w_2, w_1, w) \\ & \left. + (q)_i (p+q)_d P_{cj}(q) D_v(-q, w_2, w_1, -w_2) \right\} P_{db}^\perp(q), \end{aligned} \quad (21)$$

$$\begin{aligned} c_2 \Rightarrow & q^2 p^2 P_{ij}^\perp(p) P_{bc}^\perp(q) \left\{ (-q)_i (p-q)_c D_d(-q, w_2, w_1, -w_2) \right. \\ & \left. + (-p)_c (q-p)_i D_d(-p, w_1, w_2, -w_1) \right\} \times P_{db}^\perp(q). \end{aligned} \quad (22)$$

5.2.3 Building blocks c' and c''

$$c' = P_{ij}^\perp(p) P_{bc}^\perp(q) \left\{ -i[(q)_i \delta_{cd} + (p)_c \delta_{id}] \right. \\ \left. - (p+q)^2 \left[- (p)_c P_{id}^\perp(p) A_v(p, w_1, w_2) - (q)_i P_{cd}^\perp(q) A_v(q, w_2, w_1) \right] \right\}, \quad (23)$$

$$c'' = P_{de}^\perp(p+q) \left\{ i(q)_j \delta_{ef} - (p)^2 (q)_j P_{ef}^\perp(q) A^\nu(-q, -w_2, w_1 + w_2) \right\} P_{fb}^\perp(q). \quad (24)$$

5.2.4 Coefficients c_1, c_2, c_3 in $\text{Tr}[\Gamma^{(3)}]$

$$c_1 = c' \cdot c''. \quad (25)$$

$$c_2 = c' P_{de}^\perp(p+q) \frac{(p+q)^2}{(p)^2} \left\{ p_e P_{jf}^\perp(p) A_D(-p, -w_1, w_1 + w_2) \right\} P_{fb}^\perp(q). \quad (26)$$

$$c_3 = P_{ij}^\perp(p) P_{bc}^\perp(q) \left\{ \frac{(p+q)^2}{q^2} \left[(q)_i P_{cd}^\perp(q) A^D(-q, w_1 + w_2, -w_2) \right. \right. \\ \left. \left. - (p+q)_i P_{cd}^\perp(p+q) A^D(p+q, w_1 + w_2, -w_2) \right] \right\} \cdot c''. \quad (27)$$

5.3 Equations for (2,0)

5.3.1 Structure of the traces

$$\text{Tr}[\Gamma^{(4)}] = \partial_\kappa R_\kappa c (\bar{G}_{u,\bar{u}}^2 + G_{u,\bar{u}}^2). \quad (28)$$

$$\text{Tr}[\Gamma^{(3)}] = \partial_\kappa R_\kappa (c_1 \bar{G}_{u,\bar{u}}^2 + c_2 G_{u,\bar{u}}^2). \quad (29)$$

5.3.2 Coefficient c in $\text{Tr}[\Gamma^{(4)}]$

$$c = P_{ij}^\perp(p) P_{bc}^\perp(q) q^2 \left\{ p_c (p+q)_j P_{id}^\perp(p) D(-p, w_1, w_2, -w_1) \right. \\ + q_i (p+q)_j P_{cd}^\perp(p) D(p, w_2, w_1, -w_1) \\ + (-p)_c (q-p)_i P_{jd}^\perp(p) D(p, -w_1, w_2, w_1) \\ \left. + q_j (q-p)_i P_{dc}^\perp(p) D(p, -w_2, -w_1, w_1) \right\} P_{db}^\perp(q). \quad (30)$$

5.3.3 Building block c'''

$$c''' = P_{ij}^\perp(p) P_{bc}^\perp(q) \times \left\{ -i(q)_i \delta_{cd} + (p)_c \delta_{id} \right. \\ \left. - (p+q)^2 \left[- (p)_c P_{id}^\perp(p) A^v(p, w_1, w_2) \right] - (q)_i P_{cd}^\perp(q) A^v(q, w_2, w_1) \right\} \times P_{de}^\perp(p+q). \quad (31)$$

5.3.4 Coefficient c_1

$$\begin{aligned}
c_1 = & P_{ij}^\perp(p) P_{bc}^\perp(q) \times \left\{ -i(q)_i \delta_{cd} + (p)_c \delta_{id} \right. \\
& - (p+q)^2 \left[- (p)_c P_{id}^\perp(p) A^v(p, w_1, w_2) \right] - (q)_i P_{cd}^\perp(q) A^v(q, w_2, w_1) \left. \right\} \times P_{de}^\perp(p+q) \\
& \times \left\{ -i[(p+q)_j \delta_{ef} + (-p)_e \delta_{fj}] \right. \\
& \left. - (q)^2 \left[(p)_e P_{fj}^\perp(p) A^v(-p, -w_1, w_1 + w_2) - (p+q)_j P_{ef}^\perp(p+q) A^v(p+q, w_1 + w_2, -w_1) \right] \right\} \times P_{fb}^\perp(q).
\end{aligned} \tag{32}$$

This factorizes as $c_1 = c''' \times (\text{remainder})$, where c''' is the first bracketed factor identified above.

5.3.5 Coefficient c_2

$$\begin{aligned}
c_2 = & c''' \times \frac{(q)^2}{(p+q)^2} \times \left\{ (q)_j P_{ef}^\perp(q) A_D(-q, w_1 + w_2, -w_2) \right. \\
& \left. - (p+q)_j P_{ef}^\perp(p+q) A_D(p+q, w_1 + w_2, -w_2) \right\} \times P_{fb}^\perp(q).
\end{aligned} \tag{33}$$